MIL-C-43522A

2 May 1973
SUPERSEDING
MIL-C-43522
4 August 1967

MILITARY SPECIFICATION

CAN, RADIATOR FILLING AND CAN SPRINKLING; PLASTIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for plastic water cans.

1.2 Classification. The plastic cans shall be of the following types as specified (see 6.2 and 6.3).

Type I - 12-quart with spout
Type II - 12-quart with sprinkling head

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-390 - Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium and High Density)
L-P-403 - Plastic Molding Material, Polytetrafluoroethylene (TFE-Fluorocarbon)
PPP-B-636 - Boxes, Shipping, Fiberboard

FSC 7240

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

* National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

* Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The plastic water cans shall be the manufacturer's standard commercial design, except as specified herein.

3.2 Material.

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3.2.1 Polyethylene. The plastic material shall be virgin polyethylene conforming to type I, class L or M, grade 1 or 2 of L-P-390. Clean, unburned, plastic material in the form of imperfect parts, sprues, runners or other scrap of the same composition as the virgin material and produced in the molding or finishing operation may be reground and mixed with the material specified. Items made by rotational molding shall be made of virgin material only.

* 3.3 Design and construction.

* 3.3.1 Type I. The type I water can body, handle and spout shall be made of the material specified in 3.2.1. The type I water can shall be large enough to hold its specified capacity of water. The can shall have one or more handles that can be used for carrying and pouring. The can shall have a pouring spout long and high enough so that no spillage occurs as specified in 3.5. The end of the pouring spout shall be not more than 1 inch in diameter. The plastic can shall conform to the requirements specified in table I.

* 3.3.2 Type II. The type II water can body, handle and spout shall be made of the material specified in 3.2.1. The type II water can shall be constructed the same as type I except that it shall be equipped with a sprinkler head not less than 3-1/4 inches in diameter and having a wall thickness of 0.075 inches minimum. The sprinkler head shall be removable for cleaning without the use of tools. The water can shall conform to the requirements of table I.

TABLE I. Dimensions (inches) and capacities

Type	Capacity 1/ (nominal)	Wall thickness (minimum)
I	12-Qts.	0.075
II	12-Qts.	0.075

1/ Nominal denotes ± 5% of indicated values.

3.4 Capacity and leakage. The water cans shall contain the specified quantity of water (see table I) and shall show no evidence of leakage when tested as specified in 4.2.2.3.

3.5 Spillage. The water cans shall not spill water when tested as specified in 4.2.2.3.

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* 3.6 Rigidity. The water cans shall maintain their rigidity when tested as specified in 4.2.2.3.

3.7 Handle test. The handles of the water cans shall show no breakage or distortion when tested as specified in 4.2.2.3.

* 3.8 Drop test. The water cans shall show no breaks or cracking when tested as specified in 4.2.2.3.

3.9 Marking for identification. The outside of the water cans shall be marked for identification with the manufacturer's name or trademark.

* 3.10 Color. Unless otherwise specified (see 6.2), the color of the water cans shall be as regularly furnished by the supplier.

* 3.11 Finish. The finished water cans shall have a uniform finish in accordance with the suppliers regular commercial practice.

* 3.12 Workmanship. The finished water cans shall conform to the quality and grade of product established by this specification and shall be free from blisters, mold marks, weld marks, cracks, lumps and other defects that will affect the appearance or utility of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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* 4.2.1.1 Certificates of compliance. Materials may be accepted on the basis of a supplier's certificate of compliance for requirements specified in 3.2.1. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2.2 Inspection of the end item. The lot shall consist of all finished water cans of the same type offered for inspection at one time. The sample unit shall be one finished water can.

* 4.2.2.1 Visual examination. The finished water can shall be examined for defects listed below. The inspection level shall be II and the acceptable quality level (AQL) shall be 2.5, expressed in terms of defects per hundred units.

Examine	Defect
Color	Not as specified. Chalked, burned, bloomed or scaly.
Construction and workmanship (type I and II)	Plastic cans not free of fins and flash, rough or sharp edges. Not smooth and uniform in texture. Warped, twisted, malformed, crazed, pitted, bubbled, heat marked, blistered or uneven suface. Handle or handles missing. Handle or handles not free of fins and flash, rough or sharp edges. Handle twisted or malformed. Spout twisted or malformed. Spout not free of fins or flash, rough or sharp edges. Sprinkler head not removable by hand (when applicable).
Marking for identification	Manufacturer's name or trademark missing. Manufacturer's name or trademark incomplete or illegible.

* 4.2.2.2 Dimensional examination. The water cans shall be examined for defects in dimensions. Any dimension that does not conform to the requirements specified in table I shall be classified as a defect. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

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4.2.2.3 Testing of the end item. The finished water cans shall be tested in accordance with table II, for the characteristics indicated therein. The inspection level shall be S-2 and the AQL shall be 2.5, expressed in terms of defects per hundred units. Any evidence of noncompliance with requirements shall constitute a defect.

TABLE II. Testing of the end item

Characteristic	Requirement paragraph	Test method	Requirements applicable to sample unit	Number of determinations per sample unit	Results reported as
Capacity	3.4	4.3.2	X	1	Pass or fail
Leakage	3.4	4.3.1	X	1	Pass or fail
Spillage	3.5	4.3.1	X	1	Pass or fail
Rigidity	3.6	4.3.1	X	1	Pass or fail
Handle test	3.7	4.3.3	X	1	Pass or fail
Drop test	3.8	4.3.4	X	X	Pass or fail

* 4.2.3 Examination of preparation for delivery requirements. An examination shall be made to determine that the packaging, packing and marking complies with section 5. Defects shall be as specified below. The sample unit shall consist of one shipping container fully prepared for delivery with the exception that it shall not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5, expressed in terms of defects per hundred units.

Examine	Defect
Markings	Incorrect, incomplete, illegible; omitted, or improper size, location, sequence or method of application.
Materials	Any nonconforming component; component missing or damaged.
Workmanship	Inadequate application of components; such as incomplete closure of container flaps, improper taping, loose or inadequate sealing, strapping or stapling. Bulged or distorted container.
Contents	Number of plastic water cans per shipping container is more or less than required.

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4.2.3.1 Examination for palletization. When applicable, an examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be as listed below. The sample unit shall consist of one palletized unit load fully prepared for delivery. The lot shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 expressed in terms of defects per hundred units.

Examine	Defect
Finished dimension	Length, width or height exceeds specified maximum requirements.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

4.3 Test procedures.

4.3.1 Leakage, rigidity and spillage tests. The water can, as applicable, shall be filled to capacity with water. The can shall retain the water, without leakage, for 48 hours. The can shall be sufficiently rigid so that there shall be no perceptable distortion when filled with water. Furthermore, the cans shall be capable of being tilted to pour out the contents without noticeable distortion or difficulty from the tendency of the side walls to distort. When cans, filled with water, are tilted toward the spout to an angle of 30° there shall be no spillage at the spout or sprinkling head areas (see 3.4).

4.3.2 Capacity test. The cans, after completion of tests in 4.3.1, shall be emptied into a standard measure to determine compliance with the capacity requirements of table I (see 3.4).

4.3.3 Handle test. The water can shall be loaded with a weight equivalent to twice the weight of the nominal capacity of water. The can shall then be lifted by the handle to a position of normal use. The handle and points of attachment to the body of the can shall be examined to determine breakage or distortion (see 3.7).

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4.3.4 Drop test. The water can, when empty, shall be dropped 3 times from a height of 2 feet on a flat steel plate, concrete or macadam surface and then filled with water. The can shall show no evidence of being broken or cracked and shall not leak when filled with water and allowed to stand for at least 10 minutes (see 3.8).

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Four cans of one type only shall be packed in the most compact manner within a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each fiberboard container shall be fitted with a taped box liner conforming to type CF, class weather resistant, variety DW, grade V15c of PPP-B-636. Each fiberboard container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced in accordance with the appendix of PPP-B-636.

* 5.1.2 Level B. Four cans of one type only shall be packed in the most compact manner within a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each fiberboard container shall be fitted with a taped box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification.

* 5.1.2.1 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

* 5.1.3 Level C. Cans shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 5.2 Palletization. When specified, (see 6.2), cans of one type only, packed as specified in 5.1, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

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* 5.3 Marking. In addition to any special marking required by the contract or order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type I water can is intended for use in filling the radiators of military vehicles. The type II water can is intended for use in sprinkling water on small planted areas within military compounds.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type can required (see 1.2).
(c) Color, when applicable (see 3.10).
(d) Selection of applicable levels of packing (see 5.1).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.1.2.1).
(f) When palletization is required (see 5.2).

* 6.3 Deletion of type III. The type III - 15 gallon can with spigot covered in MIL-C-43522, dated 4 August 1967, has been deleted in this revision since there is no known FSN or requirement for the item.

* 6.4 Marginal notations. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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